

**Szegedi Tudományegyetem
Informatikai Intézet**

Skeletonization with Deep Learning

Szakdolgozat

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Feladatkiírás

A vázkijelölés 2D-ben a szegmentált bináris objektumok középvonalának meghatározását jelenti az alábbi kritériumok teljesülése mellett:

- a középvonal az objektum közepén helyezkedjen el,
- reprezentálja a kiindulási objektum általános alakját (vagyis tartalmazzon minden lényeges ágot és mentes legyen a nemkívánatosaktól),
- legyen egy pixel vékony és
- őrizze meg a kiindulási objektum topológiáját (topológia-megőrző redukcióval jöjjön létre).

A vázkijelölésre számos módszert javasoltak (pl. a távolság-transzformáción alapulókat, a kontúr mintavételezett pontjai Voronoi-diagramjának belső éleit meghatározókat és a tűzfront-terjedést szimulálókat).

A kitűzött feladat a konvolúciós neurális hálózatok (CNN), különösen a U-net architektúrák alkalmazhatóságának vizsgálata publikus képi adatbázisokon.

Abstract

This thesis aims to examine the capabilities of convolutional neural networks particularly the U-Net architecture applied to the task of skeletonization using public datasets of binary images. This was reached by building custom U-net models and combining them with different loss functions. The code for the experiments was built using Python and deep learning implementations were done using the PyTorch, MONAI and TensorBoard libraries. A public dataset collection was used for training and evaluating model performance. Custom implementations of U-Net were built. In the experiments Residual U-Net (RU-Net) and U-Net with batch normalization (BU-Net) reached the best Dice Scores. RU-Net reached 0.825 Dice score tested on unseen images from the Animal2000 dataset it was trained on, whilst BU-Net had the best generalization capabilities when tested on datasets it has not been trained on. Visual analysis of model outputs revealed that Focal Dice Loss, although often had lower Dice scores, qualitatively outperformed the other loss functions that were examined (except when paired with the Attention U-Net model).

keywords: *deep learning, skeletonization, U-Net, residual learning, attention mechanisms*

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Introduction

Skeletonization extracts an object's compact line-like feature (Ghanem and Anani 2021, 1). Geometrically speaking it's about extracting the medial axis (symmetry axis) of an object with well-defined boundaries (Menten et al. 2023; Vargas et al. 2024, 257). Importantly this method "compresses" the object while still retaining its global, local geometric and topological features (Ghanem and Anani 2021, 2). Traditional skeletonization algorithms rely on hand-crafted features, morphological techniques, and geometric relationships (Nyugen et al 2021, 1). These classic methods, however, are sensitive to noise around the boundaries and slight changes in the object's shape or contour often resulting in skeletons with too many or too few branches (Nguyen 2021, 106). In the case of thinning ensuring the preservation of topology is challenging (Menten et al 2023, 214). Furthermore, traditional algorithms lack the semantic knowledge to be able to reliably extract skeletons from natural images with multiple objects (Jerripothula et al. 2021, 14).

Subsequently, the usage of deep learning methods for image processing has become more reliable. In recent years there have been great leaps in semantic segmentation using deep learning. Skeletonization can be considered a sub-problem of image segmentation as it is often modeled as a pixel based binary classification problem – in other words a semantic segmentation problem (Nyugen et al 2021, 107)). This raises the question of how well we can adapt solutions from semantic segmentation to the more specific problem of skeletonization. For skeletonization the neural network must classify every single point whether it belongs to the skeleton or not (Ko et al 2021, 89). However, there are a few key challenges to this approach: how do we ensure that the model outputs a one-pixel wide skeleton (Ko et al 2021, 80); preserves the object's topology and the continuity of the skeleton. Additionally, it is challenging to find accurate metrics to assess model performance during training (loss functions) and model evaluation (accuracy metrics).

The goal of this thesis is to assess the capability of deep learning networks on binary image skeletonization. As per the scope of this thesis and hardware limitations, only small binary images are used, however it is important to note that deep learning methods exist for skeletonization on noisy in-the-wild pictures that traditional algorithms can't handle without manual image processing.

The models for this thesis are built on the U-net architecture and different mechanisms were added to it to see if these methods could increase model performance. U-Net was chosen because currently it is one of the most widely used state-of-the-art architecture for semantic

segmentation and fundamentally its fundamental task is to produce reduced versions of the input images which align well with the concept of skeletonization (Ghanem 2021, 1).

Besides the U-Net architecture and its variants, different loss functions were also tested. Finding the optimal loss function for skeletonization is a tricky task due to three distinct challenges: extreme class imbalance (between background pixels and pixels belonging to the skeleton); the need for topological preservation and finally ensuring a one-pixel wide continuous skeleton for the output. Thus, the goal is to test different loss functions with different U-Net variations to see if they could overcome these challenges.

1. LITERATURE REVIEW

1.1. *Deep learning architectures for segmentation*

The first major shift and breakthrough for segmentation came from Long et al.'s 2015 paper which introduced the first fully convolution network (FCN) that extended previous classification networks and made pixel-level predictions possible. They also introduced skip connections to avoid the loss of global information (Long et al. 2015). U-Net improved upon the design of FCN and reduced the number of training samples need whilst still achieving comparable segmentation results (Ronneberger et al. 2015, 2; Figure 1). U-Net became the foundation for many of the modern state-of-the-art segmentation models.

U-Net however suffers from crucial constraints: network depth is limited due to the problem of vanishing gradients that happens if the network gets too deep (Li et al. 2022, 4), it still lacks the ability to retain global context failing to segment complex images (Petit et al. 2021, 2), furthermore U-Net struggles in the case of extreme class imbalance, which is a major hurdle in skeletonization. These shortcomings necessitate further improvements on the architecture.

In the same year (2015) that U-Net released Google's research team introduced another important architectural design called ResNet, which used shortcut connections to address the degradation problem (decreasing learning rate with model depth) in extremely deep neural networks (He et al. 2015, 8).

Fully convolutional neural networks, skip connections (from U-Net); shortcut connections (from Res-Net) and the U-shaped architecture became foundational concepts that together influenced the direction of modern image processing with deep learning. The joint integration of concepts behind these models have collectively defined the direction of segmentation research.

Challenges with generating and retaining fine details (like skeletons), and vanishing gradients pushed researchers to explore other potential solutions such as GANs (generative adversarial network) and recurrent-residual networks. SkelGAN introduced in 2021 used GANs with a discriminator to solve the font skeletonization problem (Ko 2021, 4), whilst the authors of ResUnet used shortcut connections and in R2U-Net added recurrency (Zhang et al. 2018, Alom et al. 2019). The authors of SkelGAN emphasize that the lack of available perfect ground truth labels inhibits quantitative evaluation, which as we will see is an important concern in

skeletonization, and their model often generates artificial features such as noisy edges and broken lines (Ko 2021, 7-8). In R2U-Net introducing recurrent layers led to significant increase in training time (Alom et al. 2019, 13), thus leading to a trade-off between learning time and performance improvements based on the given task the model aims to solve. Because these methods increase computational costs and visual errors remain the next section examines another type of solution: attention mechanism.

To curb computational needs (overhead) and avoid the need for multi-stage processing chains attention mechanisms were introduced into the segmentation architectures. Attention U-Net model used AGs (Attention Gates) to increase the model's prediction accuracy by automatically suppressing irrelevant background regions, whilst highlighting relevant objects without increasing computational costs (Oktay et al. 2018, 1-2). This successfully eliminated the need for external localization models (Oktay et al. 2018, 8), even this enhanced U-Net with attention mechanism was only able to reach 0.62-0.68 F1 score – which is one of the standard metrics for deep learning segmentation, more on this in later chapters – on the skeletonization task, since this architecture is not optimized for topological structures (Nguyen 2021, 2108). To resolve this limitation attention mechanisms, have to be combined with other learning solutions.

Modern segmentation models use hybrid task specific architectures combining multiple mechanisms concurrently such as the DAttRes-Unet, Residual-Attention UNet++ or auxiliary learning task models. Auxiliary task learning, TTA (Test Time Augmentation) and dynamic threshold finding together achieved 0.80 F1 Score (Nguyen 2021, 2105-8), whilst DAttRes-Unet and the Residual-Attention UNet++ that combined attention gates and residual blocks successfully mitigated the degradation problem (Li et al. 2022, 4; 2022 1). Despite the improvements these hybrid approaches brought a critical limitation remains: the lack of a large, labeled training dataset (Li et al. 2022, 2), and though model complexity is increased these approaches still only yield incremental improvements (often only 1-2 percentages).

Other approaches diverge from the CNN approach completely using more complex domain-agnostic algorithms. SkeletonNetV2 uses dense channel attention blocks (DCAB), multi-level supervision, pruning to create an end-to-end network that reaches 0.7961 F1 score (Nathan and Kansal 2021). The Skellite model was optimized for extracting thin features using a synthetic training dataset and achieved without retraining good generalizability on new datasets, whilst increasing speed and ensures one-pixel wide skeletonization outputs (Vargas et al. 2025, 1, 9-10, 12). Menten et al. (2023) uses a differentiable mathematic approach to systematically remove background pixels whilst ensuring perfect connectivity without the usage of training data.

Whilst these are innovative methods, they remain limited: Skellite was trained and evaluated almost exclusively on thin curvilinear and synthetic data, which raises the question if it could perform efficiently on complex, thick real-life objects, whereas the algorithm introduced by Menten et al. is significantly slower to process data than predictive models like the U-Net.

These compromises and limitations discussed in this section highlight that architectural innovation alone is not enough to reach optimal performance; the performance of models and algorithms is also influenced by other factors such as dataset quality, available loss functions and validation metrics.

1.2. Datasets, Loss Functions and Metrics

The quality of machine learning is fundamentally determined by the quality of datasets used for training and evaluation and the adequacy of loss function and validation metrics for the given task. The lack of high quality, robust benchmark datasets is a major limiting factor for trying to solve skeletonization with deep learning. The underlying cause is that it's hard to generate high quality ground truth masks (skeletons). The SkeView dataset aims to close this gap by using a human-in-the-loop approach to create a high-quality training and benchmark dataset making it possible to compare models more objectively (Yang et al. 2024).

Though creating standardized high quality benchmark datasets is a step towards better benchmarking, however the lack of sufficient standard evaluation metric also causes issues when trying to compare different models. F1 and Dice scores or the Jaccard-index (also called IoU for Intersection over Union) are segmentation metrics that when used for skeletonization do not offer adequate performance validation as these metrics lack the ability to measure the key features of skeletons: topology preservation and thin 1-pixel wide output. Thus, for skeletonization quantitative performance analysis is necessary. Though the metrics mentioned are flawed for evaluating skeletonization still research should emphasize including standard metrics, alongside other statistics such as training time (with exact hardware parameters), and as many visual examples as possible for easy comparison between models and reproducibility. Unfortunately, many studies fail to include one or more of these, making it hard to evaluate the state-of-the-art.

Since currently there is no adequate standard metric to evaluate skeletonization, loss functions – which are often created by creating a differentiable version of an already existing metric – also lag behind. Jadon (2020) surveyed 15 different loss functions for segmentation, and their table helps researchers identify what could be the most appropriate loss function for

the task they aim to solve, future skeletonization research should create similar surveys for loss functions and metrics for skeletonization.

Currently Dice loss remains one of the most used loss functions for segmentation, though it has its limits as Jadon (2020) notes that due to its “jagged nature” it often leads to getting stuck in local minimums during training. However fundamentally, Dice lacks the capability to measure skeletonization features, and minimizing Dice Loss alone might not yield the highest results. To address this centerline Dice Loss (clDice) was created by Shit et al. (2021) specifically for retaining tubular structures’ topology and connectivity. Although clDice could help to optimize skeletonization networks it remains to be seen if it helps to ensure thin one-pixel wide skeletons. Furthermore, an important question to ask is whether for a given task ensuring thin skeletons is necessity or a possible trade off for faster implementation and training and reducing computation costs. Future research should examine the possibilities for finding a good loss metric for skeletonization and the trade-offs between following the strict rules of skeletonization and real-world applications.

1.3. 3. Goals

The goal of this thesis is to assess the capability of deep learning algorithms on binary image skeletonization by testing the main state-of-the-art mechanisms on the SkeView dataset collection with different loss functions to survey the performance of these approaches.

As per the scope of this thesis and hardware limitations, only small binary images are used, however it is important to note that many state-of-the-art methods perform well on noisy in-the-wild pictures that traditional algorithms cannot handle without manual image processing.

2. METHODOLOGY

2.1. Model architectures

2.1.1. The baseline model: U-Net

Thanks to its robust yet simple architecture, fast inference times and efficiency U-Net became a widely adopted standard baseline for image segmentation (Zhou et al. 2018, 5), including skeletonization. Since this thesis approaches skeletonization as a sub-problem of semantic segmentation U-Net provides a fast and standard baseline.

The fundamental strength of the U-Net architecture lies in its symmetric U-shape (Figure 2.1). It features an encoder path that down samples the images extracting feature maps along the way; the decoder path which up samples the feature maps making pixel-wise segmentation possible and the bridge or bottleneck that connects these two paths. The revolutionary aspect of this model lies in the skip connections that take the higher resolution feature maps and concatenate them to the coarser feature maps of the decoder path, enabling the model to see higher resolution global spatial data in the deeper layers. Making feature localization and the preservation of fine details easier.

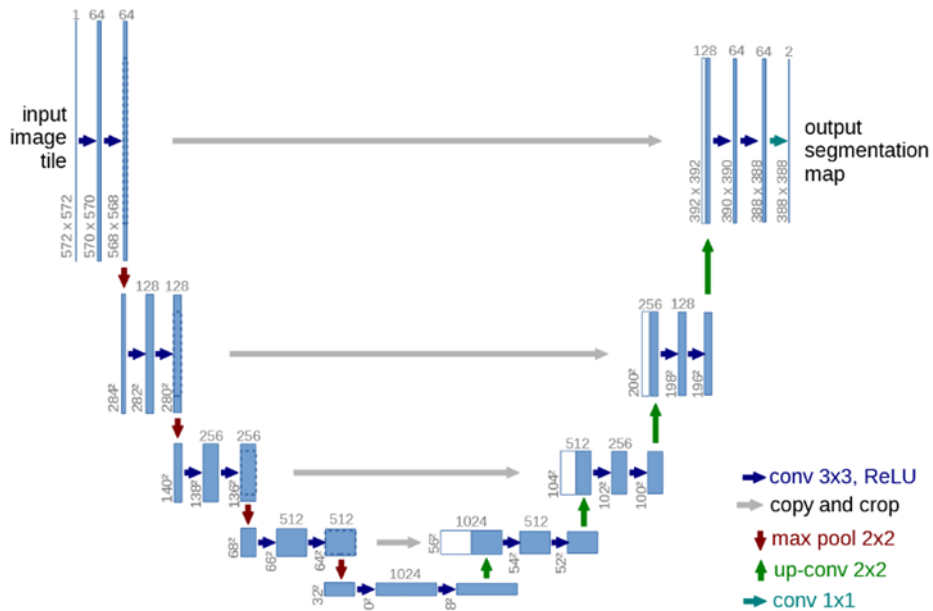


Figure 2.1 The original U-Net architecture diagram from Ronneberger et al. (2015, 2). The blue boxes are feature maps, white boxes signify copied feature maps (Ronneberger et al. 2015, 2). Arrows mark operations (Ronneberger et al. 2015, 2). This model's implementation with slight modifications is used as a baseline for the model, as well as for the foundation of the main model for the thesis.

The thesis uses custom implementations based on the original U-Net model from the paper by Ronneberg et al. (2015). For the baseline U-net model although the original architecture used cropping to address the spatial discrepancy between the skip connection feature maps and the decoder path, the implementation used for this thesis uses padding in the convolutional blocks to ensure spatial dimensions are preserved and thus the skip connection feature maps can be concatenated with the corresponding decoder block's feature map directly without information loss.

Despite its success U-Net struggles to solve tasks with extreme class imbalance (like skeletonization) and vanishing gradients. To address these challenges the next section introduces a hybrid architecture based on research to test if using residual learning and attention mechanisms would improve performance on skeletonization.

2.1.2. Residual Attention U-Net (RAU-Net)

Training deep neural networks often faces the challenge of degradation and vanishing gradients. Degradation means that the increasing model depth often leads to a decrease in accuracy (He et al. 2015, 2). Vanishing gradients occur when the backward propagated gradients diminish as they pass through multiple layers (He et al. 2015, 2; Li et al. 2022, 4). Residual learning aims to address these limitations by using identity mappings and residual blocks, which use parameter-free “shortcut connections” to pass on information (He et al. 2015, 2-3). Integrating this architecture into segmentation models ensures that signal information propagates without degradation both in low- and high-level features, making the networks easier to train (Zhang et al. 2018, 3).

Therefore, residual blocks (Figure 2.2) were added to the custom U-Net model built for this thesis. These blocks were based on the paper of He et al. (2015). Residual blocks take the feature map from the previous block and after a single convolution to scale the identity map to appropriate dimensions, an element-wise addition adds together the identity map and the output of the block, thus along with batch normalization they effectively preserve the higher-level signal, reducing the chance of vanishing gradients. Batch normalization is also used as a regularization technique to prevent the model from overfitting, as with residual learning the chances of overfitting increase as the model can learn faster and more effectively. During the experiments done for this thesis, training time converged faster with residual blocks reaching

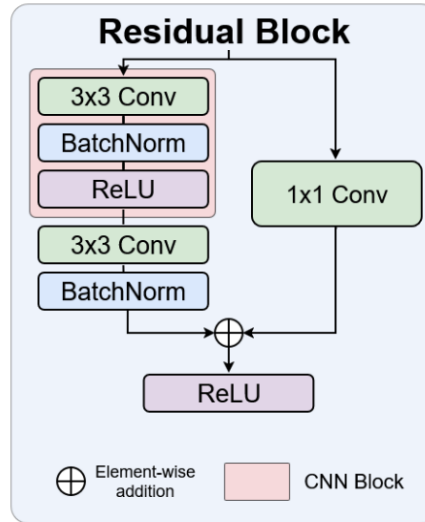


Figure 2.2 The residual blocks used in the custom RAU-Net. Boxes mark different layers, arrows show the direction data passes through the network and the circle with the cross inside it marks element-wise addition.

peak accuracy in earlier epochs than the U-Net models without residual blocks conforming that residual learning effectively decreases training times. Whilst residual blocks ensure that structural information is preserved throughout the network, attention mechanisms were introduced into the network to help the model focus on relevant information in the context of extreme class imbalance.

Attention mechanisms were through adding Attention Gates (AGs) as a gateway between skip connections (Figure 2.3). The AGs used for this thesis were based on Oktay et al.'s 2018 paper. The skip connection feature map is first sent through the attention gate and element-wise addition is used to add the coarse feature map from the deeper layer of the network to create an attention map (alpha) which in the context of the model is then used as a weighing factor to the skip connections (by element-wise multiplication) ensuring that the skip connection input the decoder block gets has been weighted based on relevancy.

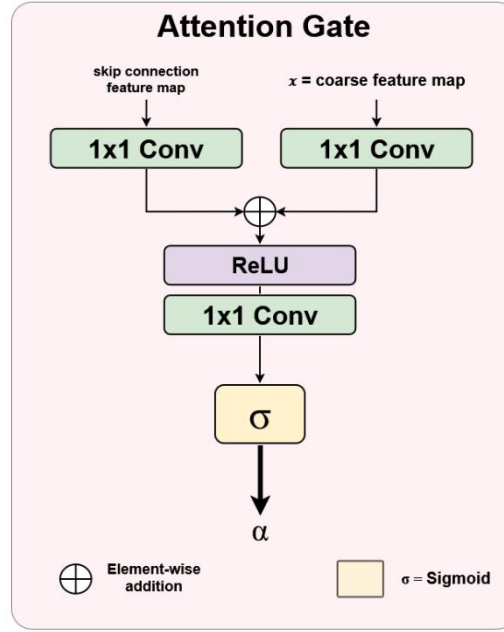


Figure 1.3 The Attention Gate block used in RAU-Net. These return the attention maps (α) used to highlight relevant areas and suppress irrelevant regions. These intercept the skip connections in the RAU-Net model. Boxes mark network layers, arrows show how data moves passes through the network and the circle with the cross inside it marks element-wise addition.

The architecture of Residual Attention U-Net (RAU-Net) (Figure 2.4) is based on the original U-Net model (Figure 2.1); however, it is enhanced by the usage of residual blocks (Figure 2.2) and attention gates (Figure 2.3). This results in theory in an architecture that maximizes information preservation and minimizes the impact of irrelevant features. In this combined model the original convolutional blocks (CNN Blocks) are changed to residual units guaranteeing network stability even as depth increases. High-resolution spatial information from the skip connections is filtered through Attention Gates (AGs) before being concatenated into lower-level feature maps (Zhang et al. 2018, 3; Oktay et al. 2018, 5). AGs using the course scale contextual information suppress noise, enabling U-Net’s decoding path to build from only the most relevant and precise features (Oktay et al. 2018, 5). This in theory is a hybrid system that creates a stable base for robust skeletonization by addressing the key limitations of the U-Net model.

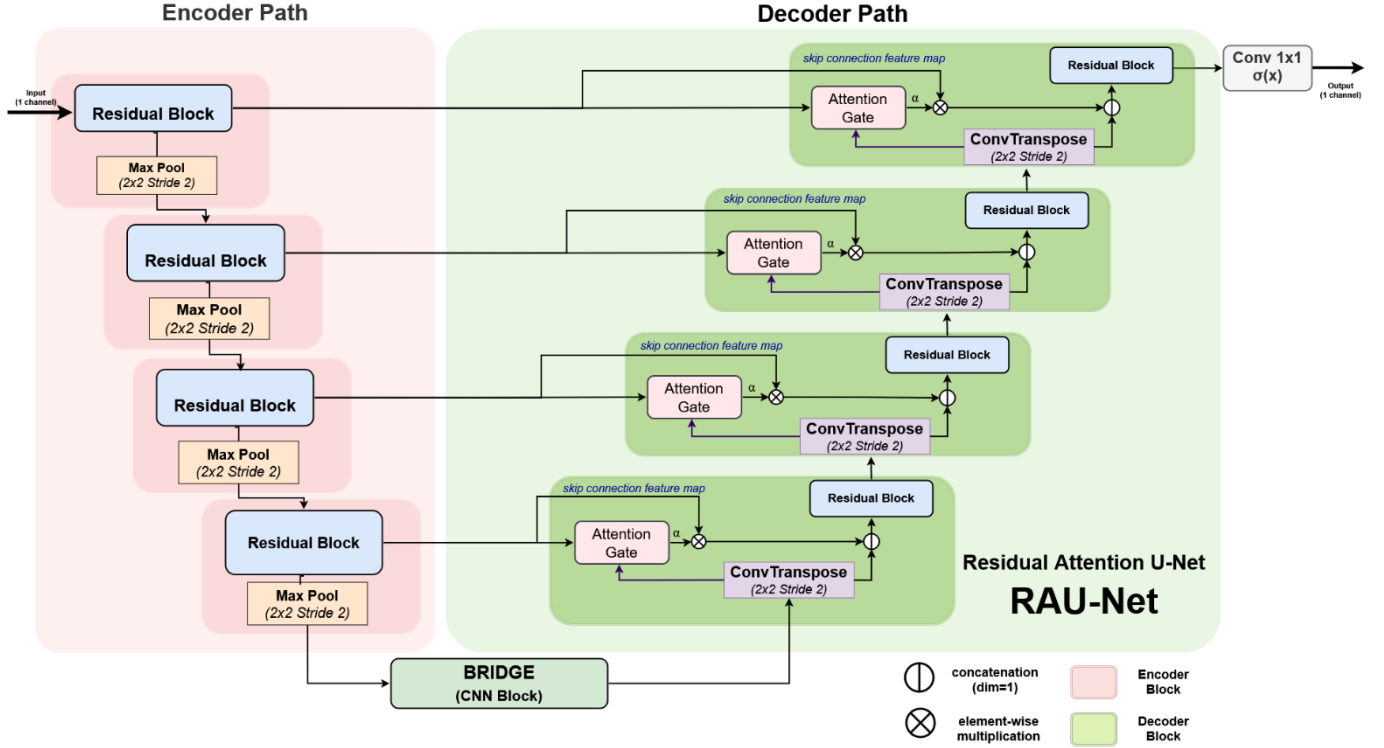


Figure 4 The architecture of the RAU-Net model. The modular implementation makes it possible to switch off the residual and/or attention mechanisms. Boxes on the figure mark network layers, arrows show the direction data passes through the network. The circle split in half signified concatenation, whilst the circle with the x inside of it marks element-wise multiplication.

To accurately test the capabilities of each mechanism the RAU-Net model was built in a modular way. Residual blocks and attention gates could be turned off separately, resulting in three other models: a U-Net network with batch normalization (BU-Net), an Attention U-Net (AU-Net) and Residual Attention U-Net (RU-Net). These are tested with the combination of different loss functions.

2.2. Loss functions and Metrics

To evaluate segmentation in deep learning, it is necessary to calculate the overlap between the ground truth masks and the predicted mask. The most widely used metrics for this are the Dice-coefficient (DICE, mathematically the same as the F1 score) (1: Taha and Hanbury 2015, 5) which also correlates to the Jaccard-index (JAC) (2: Taha and Hanbury 2015, 5), also called Intersection over Union (IoU) (Taha and Hanbury 2015, 5).

$$DICE = \frac{2TP}{2TP + FP + FN} \quad (1)$$

$$JAC = \frac{TP}{TP + FP + FN} \quad (2)$$

$$JAC = \frac{DICE}{2 - DICE} \quad (3)$$

Mathematically these two metrics strongly correlate (3: Taha and Hanbury, 5); both measure the same aspect of segmentation and result in the same ranking between models making it redundant to use both during evaluation (Taha and Hanbury 2015, 5). Thus, for this thesis Dice score was chosen as primary quantitative metric. Although a standard metric for measuring segmentation performance in deep learning models, these metrics (including IoU) inherently lack the nuances necessary to assess important skeletonization criteriums: topology preservation, one-pixel wide and continuous skeletons. Thus, to accurately assess model performance a qualitative human-in-the-loop approach using visual analyzation is crucial.

Dice score reflects the goals of segmentation well and thus it was adapted to a loss function: Dice Loss (4: Jadon 2020, 2). However, due to its non-convex nature (Jadon 2020, 4) it is prone to getting stuck at local minimums especially early in the training when the target class value is very low, making it necessary to combine with other metrics when stability is crucial.

$$DL(y, \hat{p}) = 1 - \frac{2y\hat{p} + 1}{y + \hat{p} + 1} \quad (4)$$

The addition of Binary Cross Entropy (BCE) to DICE creates a more stable loss function called Dice Loss with Cross Entropy (DiceCE) (5) which and its variations (like Combo Loss surveyed by Jadon 2020) are widely used as a stable baseline loss functions. Thus, DiceCE was used for training the baseline U-Net model for this thesis.

$$DiceCE = \alpha DICE + \beta BCE \quad (5)$$

$$FL = -\alpha(1 - p_t)^\gamma \log(p_t) \quad (6)$$

Another widely used loss to combine with Dice Loss is Focal Loss (FL) (6 from Jadon 2020, 2) which is a variation of Binary Cross Entropy (BCE) that decreases the weight of easy examples, forcing the model to learn hard examples in highly imbalanced datasets (Jadon 2020, table 2). Though Focal Dice Loss (FDL) was aimed for tumor segmentation (Wang and Chung 2018, 2) this mechanism may help to address the extreme class imbalance in skeletonization. But FDL like the other loss functions, also lacks solutions for topology preservation, ensuring

thinness and connectivity. To reach precise skeletonization loss functions need to address these aspects as well.

Centerline Dice (clDice) theoretically addresses this problem. clDice was aimed for segmentation problems with tubular structures where ensuring connectivity and topology perseverance is important (Shit et al. 2021, 1) by increasing topological sensitivity and precision (Shit et al. 2021, 2). Making clDice a good candidate for skeletonization. For this thesis multiple loss functions were compared: Dice, DiceCE, FDL and clDice to see how they influence model performance.

3. EXPERIMENTS

All experiments were run on an Asus ROG Strix G16 with Intel i7-13650HX processor running the Windows 11 operating system and with an NVIDIA GeForce 4060 laptop GPU with 8GB of VRAM limiting the available computational resources. Thus, the experiments were scaled to work with these hardware constraints.

The models and the code for the experiments were implemented using Python with a standard deep learning library PyTorch with Cuda and loss functions and metrics were used from the MONAI (Cardoso et al. 2021) library specifically made for segmentation tasks. PyTorch and MONAI were chosen because they contain highly optimized implementations of the core components needed for deep learning segmentation experiments as well as integrate well with the Cuda ecosystem for optimized training on the GPU.

The datasets for training, validation and testing were from the public SkeView dataset collection by Yang et al. (2024), because they offer a validated high-quality collection of major skeletonization datasets as opposed to an earlier approach to this thesis in which I tried to work on generating truth masks and datasets, which impaired model performance. From these Kimia99, Kimia216, MPEG400, MPEG7 and Animal2000 were chosen because they contain images of varied shapes and sizes. Unfortunately, the sizes of the images taken from the SkeView collection are varied and preprocessing was needed to make them uniform for training. This is reached by on-the-fly transformations done before feeding the dataset to the model for training or evaluation.

Because of hardware limitations, image sizes were constrained. Images that exceeded a certain threshold in either width or height were discarded before the dataset was split into training, validation and test datasets. Different thresholds were set for different datasets based on the dominant size range of the datasets and to better evaluate different model capabilities. Images from the Kimia datasets that on either dimension (height or width) exceeded 160 were discarded. For the other datasets (MPEG400, MPEG7 and Animal2000) 512px was used as a threshold as it was the maximum size that the hardware tolerated yet still minimized the number of images discarded. Thus, the models and loss functions were tested on a variety of images, resolutions and dataset sizes allowing a more comprehensive analysis of performance.

To ensure uniform image sizes for the remaining images padding which matched the edges was used, then center crop. No other transformations were applied on the images.

Training validation and test split also happened on-the-fly before each run, using a fixed random seed (42) to avoid data leakage during training and validation. The split was done with the standard ratio of 70% used for training, 15% used for validation and 15% used for testing.

Each of the model architectures (U-Net, AU-Net, RU-Net, RAU-Net), was tested on all datasets, with different loss functions. All models were tested with the loss functions: Dice, DiceCE, Focal Dice Loss (FDL). The Dice metric was used to measure their training, validation and testing accuracy. Focal loss, cIDice and Dice cIDice were discarded during preliminary experiments because they converged on low validation DICE scores after a few epochs. A random seed (42) was used for model initialization and training further ensuring reproducibility.

For complete reproducibility the codebase with the full implementation (including the data pre-processing pipeline) is available on GitHub along with the best pretrained models. Please note that the datasets from SkeView collection must be downloaded in the same folder structure as is provided in the code or the path variables need to be adjusted.

4. RESULTS

The baseline was set by the custom implementation of the U-Net with DiceCE Loss. During the validation and training runs the best U-Net architecture reached the Dice score of 0.68 on the validation dataset and 0.72 on the training set. U-Net’s average performance (using any loss function) across the training sets was 0.47 Dice score on the validation data and 0.52 on the training data, significantly lower than the other networks (Table 1).

Table 1 Model Performance During the Training and Validation Phase

model	Best Validation Dice Score				Best Training Dice Score			Number of Epochs		
	mean	min	max	std	mean	min	max	mean	min	max
AUNET	0.5953	0.213	0.7246	0.1183	0.777	0.2512	0.9956	82	31	161
BUNET	0.6112	0.369	0.729	0.0868	0.7825	0.4437	0.9567	87	56	121
RAUNET	0.6232	0.5559	0.7296	0.0602	0.772	0.59	0.9071	76.67	46	126
RUNET	0.6261	0.5735	0.7276	0.0534	0.7642	0.6565	0.887	76.67	41	136
UNET	0.4704	0.2028	0.6873	0.1931	0.5218	0.2064	0.7244	64	31	111

Unfortunately, the U-Net model proved to be highly unstable and sensitive to weight initialization due to the lack of normalization resulting in catastrophic activation collapse (outputting constant values) in most cases. In contrast all other models use batch normalization and maintained complete stability during the later testing phases. Because of this failure validation scores are used as a baseline for the test phase as well.

The model configurations which performed the best on the validation datasets were chosen for evaluation. The best performers from the training and validation phase were the models that were trained on the Animal2000 dataset and Kimia216 for the smaller resolution category. These models were then evaluated on the test dataset portion of the dataset they were trained on to measure in-domain performance. The models were also tested on unseen images from datasets that they were not trained on to measure generalization ability (cross-domain evaluation). The average test dice score (Table 2) was the average of these two evaluation types.

Table 2 Overall Model Performance on Test Datasets (including cross-domain)

model	Loss Function	Average Test Dice Score	Maximum Test Dice Sore	Minimum Test Dice Score
AUNET	DICE	0.6842	0.7716	0.5968
RUNET	FDL	0.6506	0.8036	0.5924
BUNET	DICE	0.6498	0.8036	0.5829
BUNET	DICECE	0.6373	0.7925	0.5692
RUNET	DICE	0.6371	0.8256	0.5674
AUNET	DICECE	0.6331	0.793	0.5583
BUNET	FDL	0.6319	0.8022	0.5485
RAUNET	DICECE	0.6272	0.7194	0.5977
RAUNET	DICE	0.6202	0.7106	0.5904
RAUNET	FDL	0.6185	0.7169	0.5836
RUNET	DICECE	0.6006	0.602	0.5993

RU-Net's performed the best overall on the in-domain test sets, showing strong learning capacity with the peak Dice score of 0.825 on the Animal2000 dataset (Table 3). BU-Net on the other hand proved superior for cross-domain generalization thus reaching a higher overall average score (Table 2 and 3). RAU-Net on the other hand proved to perform worse than the other enhanced models (though it still outperformed the baseline) showing that stacking attention and residual mechanisms did not lead to performance improvement on the examined datasets.

Table 3 Overall Model Performance on unseen data from the same dataset (Animal2000)

model	Loss Function	Training Dataset	Dice Score
RUNET	DICE	ANIMAL2000	0.825621
BUNET	DICE	ANIMAL2000	0.803624
RUNET	FDL	ANIMAL2000	0.803605
BUNET	FDL	ANIMAL2000	0.802158
AUNET	DICECE	ANIMAL2000	0.793041
BUNET	DICECE	ANIMAL2000	0.792485
AUNET	DICE	ANIMAL2000	0.771563
RAUNET	DICECE	ANIMAL2000	0.719445
RAUNET	FDL	ANIMAL2000	0.716872
RAUNET	DICE	ANIMAL2000	0.710649

Table 4 Overall Model Performance on unseen data from the same dataset (Kimia216)

Model	Loss function	Trained on	Dice Score
RUNET	DICECE	KIMIA216	0.602014
RAUNET	DICECE	KIMIA216	0.599211
RAUNET	DICE	KIMIA216	0.599087
AUNET	DICE	KIMIA216	0.596849
RUNET	FDL	KIMIA216	0.592429
BUNET	DICE	KIMIA216	0.591465
RAUNET	FDL	KIMIA216	0.583627
BUNET	DICECE	KIMIA216	0.570372
RUNET	DICE	KIMIA216	0.567371
AUNET	DICECE	KIMIA216	0.562069

Furthermore, visual analysis and the Dice Score metrics both prove that lower resolution small datasets are insufficient for training neural networks to solve complex tasks like

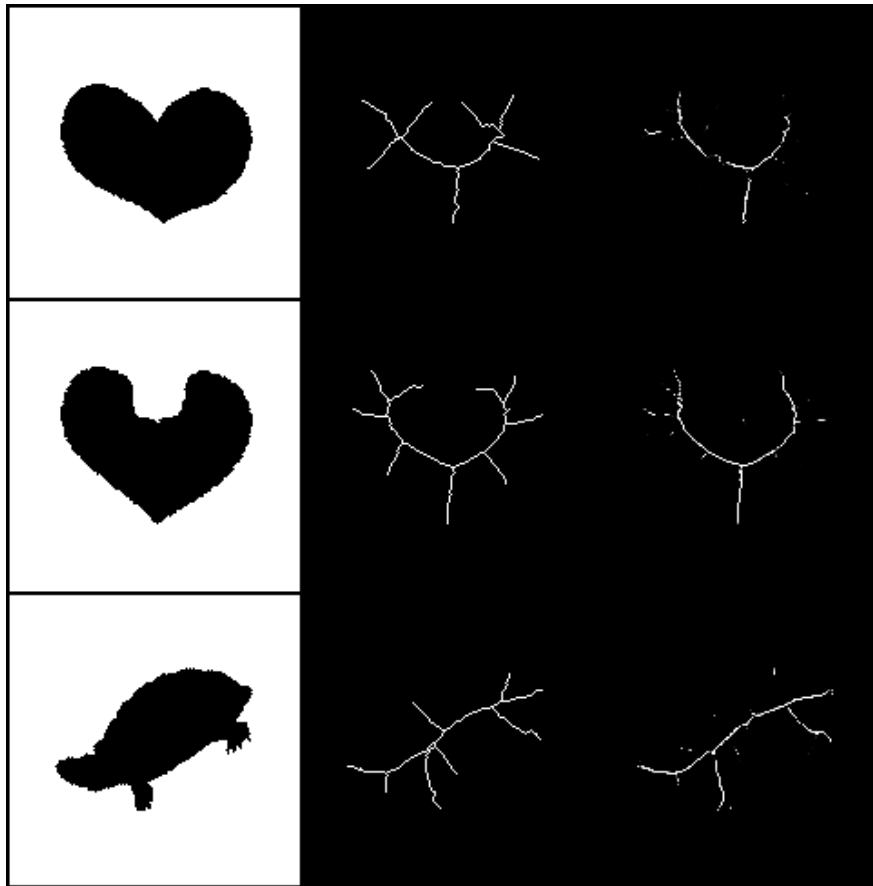


Figure 4.1 – BU-Net’s with DiceCE Loss trained on Kimia216, results on unseen images from the same dataset. (Left: original shape, Middle: ground truth mask, Right: model prediction)

skeletonization. Models trained on the Kimia dataset with low resolution (160x160) often produced broken, jittery noisy results, catastrophically failing the skeletonization task (Figure 4.1). The same model-loss setups achieved much higher quality and scores on the Animal2000 dataset (Figure 4.3).

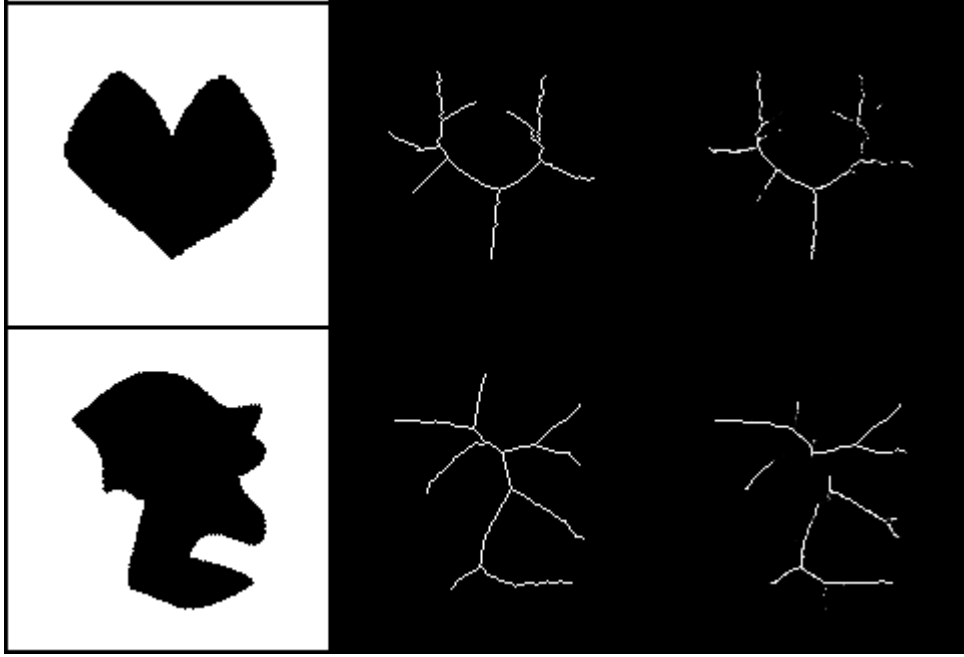


Figure 4.2 BU-Net with DiceCE loss (trained on Kimia216) inference on unseen images from Kimia216. An example of jittery noisy output. (Left: original shape, Middle: ground truth mask, Right: model prediction)

As can be seen on Figures 4.1 and 4.2 the models trained on Kimia216 generally had less confidence, which led to broken lines, lot of jitter and noise. This noise is probably due to the training datasets size (less than 200 images) which lead the models to overfit the training data and lack generalization abilities, since the same models trained on the bigger dataset (Animal2000) usually oversimplified the skeleton (Appendix Figure 4.3-4.4), but did not produce as much noise. Though the outputs are noisy for the small datasets most of the lines are often one-pixel-wide which is a key requirement in skeletonization which shows the power of overlap-based loss functions like Dice, these alone can help ensure the model does not output overly thick skeletons. The real bottleneck is ensuring topology preservation and connectivity.

Snapshots of model predictions for the validation dataset were taken during training (Appendix Figure 4.5-4.7), which shows a glimpse into how these models learn. Models using standard Dice loss were interestingly the least confident about the object's most inner parts early into training and often completely failed to classify these as part of the skeleton (Appendix Figure 4.5). As the models continued to train, they started to aggressively prune branches leading to oversimplified skeletons (Appendix Figure 4.6). Meanwhile models with Focal Dice Loss correctly identified the objects center as part of the skeleton early on (Appendix Figure 4.7), however this did not translate to unseen images in the case of small datasets, but when

trained on larger datasets (MPEG7 or Animal2000) these generalizations did lead to less breakage from models trained with Focal Dice Loss.

Models trained on the larger Animal2000 dataset consistently output a one-pixel-wide skeleton, however as mentioned before regularly oversimplified the skeleton (Appendix Figure 4.3 and 4.8). The difference between Dice Loss in terms of metric performance vs. visual performance is stark. A notable exception from this was the AU-Net architecture which thanks to its attention mechanism performed on a similar level in terms of quality even with Dice or DiceCE Losses and the usage of Focal Dice Loss did not improve its performance. Dice Loss more often led to broken lines and noise (Figure 4.8, 4.10), although it is important to note that all models with each loss type suffered from connectivity issues and some level of noise. However, compared to the models trained on Kimia216 there is a marked improvement both in the level of connectivity and noise, as well as more consistency in terms of one-pixel-wide output, meaning more and higher resolution data improved outcomes. Suggesting that using larger datasets or perhaps training models on multiple datasets and even using data augmentation like flipping and rotation could lead to even better results.

Different loss functions worked better for different architectures, showing that architecture structure also influences optimal loss function choice. Upon visual analysis of model outputs although the Dice loss function scored well in reality models trained with Dice often produced noise, jittery output even if their Dice scores remained high, meanwhile models paired with Focal Dice Loss (FDL) although often led to a slightly lower Dice score while produced cleaner results.

The case of Dice Loss function consistently outperforming Focal Dice Loss on the Dice metric, whilst it produced worse outputs highlights that Dice score as a measure of performance is inadequate for skeletonization, since the model can achieve high overlap scores without ensuring connectivity. Thus, proving the argument that there is a need for a standard metric to measure model performance on tasks like skeletonization. Creating such metric could also trickle down to loss functions improving the overall potential of deep learning models to solve the task of skeletonization.

5. CONCLUSIONS

In conclusion, skeletonization remains a challenging task for deep learning. Although enhanced modular networks outperformed the baseline U-Net model in metrics and visual outputs, these architectures could not consistently meet the fundamental strict topological requirements of skeletonization. Furthermore, the experiments conducted in this thesis supported the idea that Dice Score is an inefficient and often misleading metric for skeletonization as it fails to penalize topological errors. A key limitation for deep learning skeletonization remains the lack of large, standardized datasets. Without access to comprehensive, high-resolution data, it remains difficult to train and benchmark models accurately. To address these limitations, future research must prioritize the creation of larger, specialized datasets.

Another key limitation during the research process (besides having standardized metrics to measure performance) was the lack of details for reproducibility. Many studies omit the environments they used to conduct the experiments such as the hardware and the software used. More examples of model outputs would also be crucial for evaluation as well as access to implementations whenever possible, perhaps even pre-trained models. These would help to identify different methods and help find the best directions towards new research.

Additionally, I propose that the field requires extensive survey studies focused on loss functions, evaluation metrics and benchmarking different architectures on skeletonization. Surveys would help future researchers discover the underlying trends and create new state-of-the-art models.

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Nyilatkozat

Alulírott Nagy Szilvia Kamilla programtervező informatika szakos hallgató, kijelentem, hogy a dolgozatomat a Szegedi Tudományegyetem, Informatikai Intézetén készítettem, programtervező informatika Bsc diploma megszerzése érdekében.

Kijelentem, hogy a dolgozatot más szakon korábban nem védtem meg, saját munkám eredménye, és csak a hivatkozott forrásokat (szakirodalom, eszközök, stb.) használtam fel.

Tudomásul veszem, hogy szakdolgozatomat / diplomamunkámat a Szegedi Tudományegyetem Diplomamunka Repozitóriumban tárolja.

Dátum 2026. május. 23.

Nagy Szilvia Kamilla

Appendix

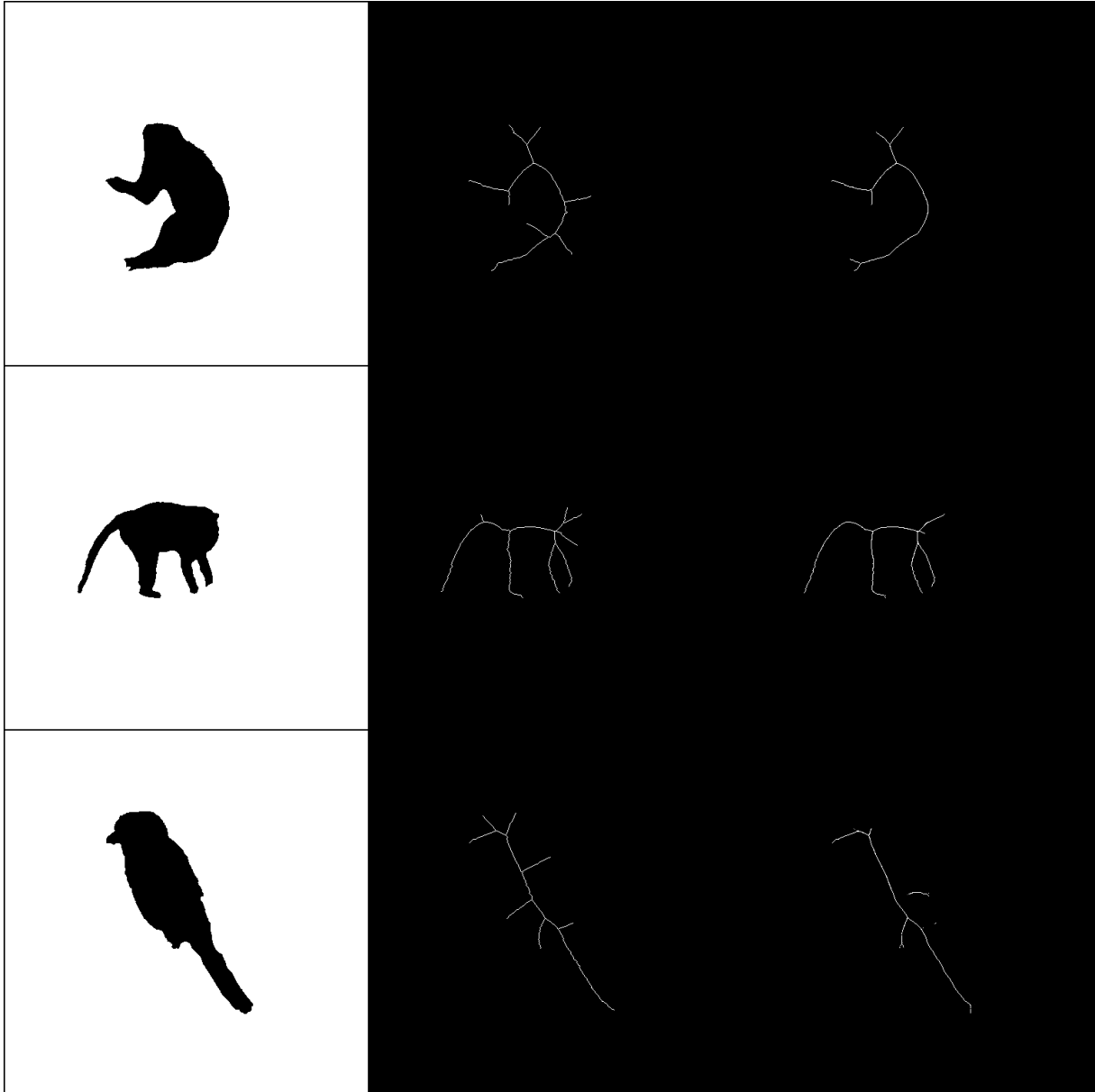


Figure 4.3 BU-Net with Dice Loss trained and tested on Animal2000. The images show the model's performance on unseen images from the same dataset. (Left: original shape, Middle: ground truth mask, Right: model prediction)

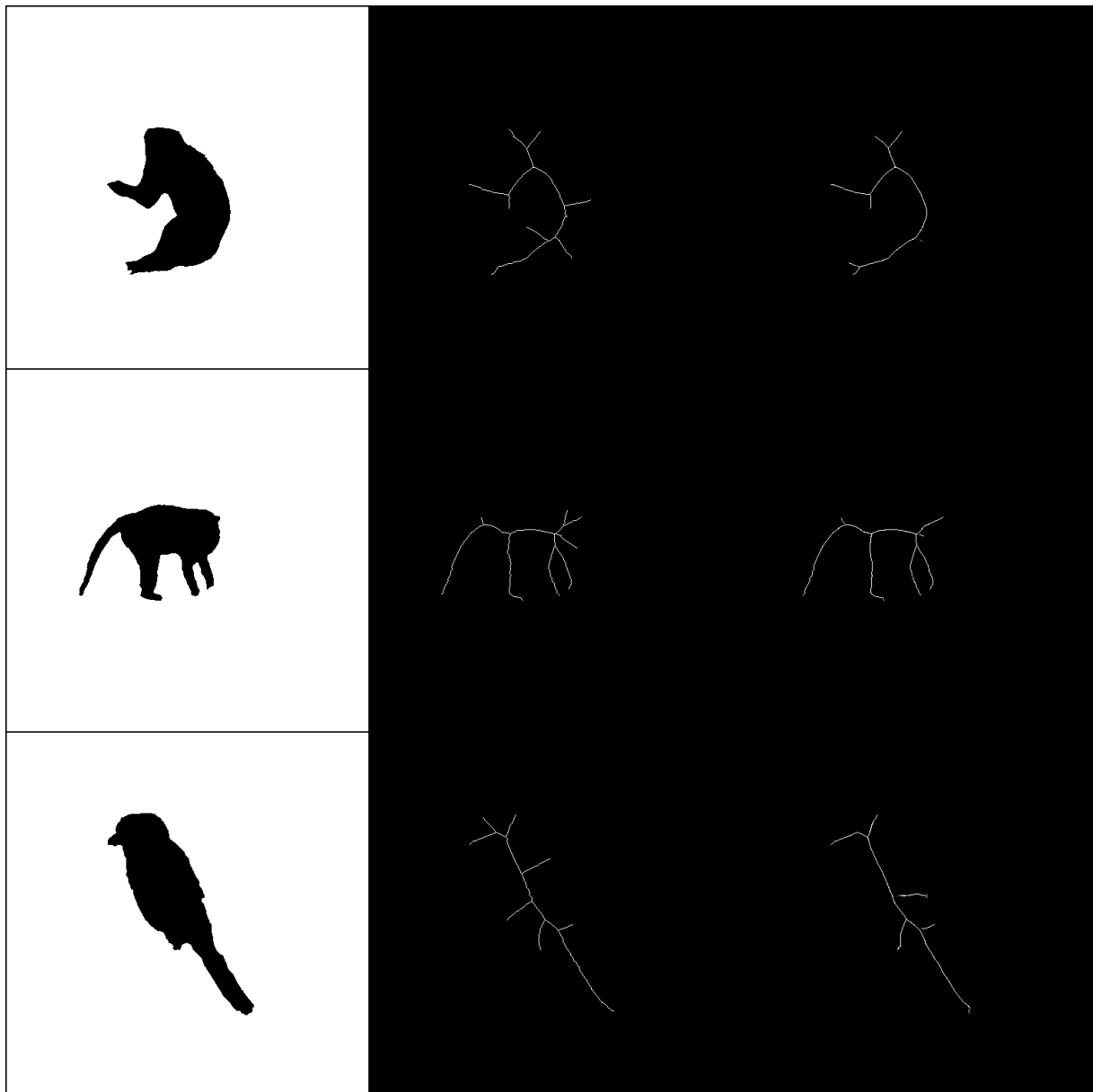


Figure 4.4 BU-Net with Focal Dice Loss trained and tested on Animal2000. The images show the model's performance on unseen images from the same dataset. (Left: original shape, Middle: ground truth mask, Right: model prediction)

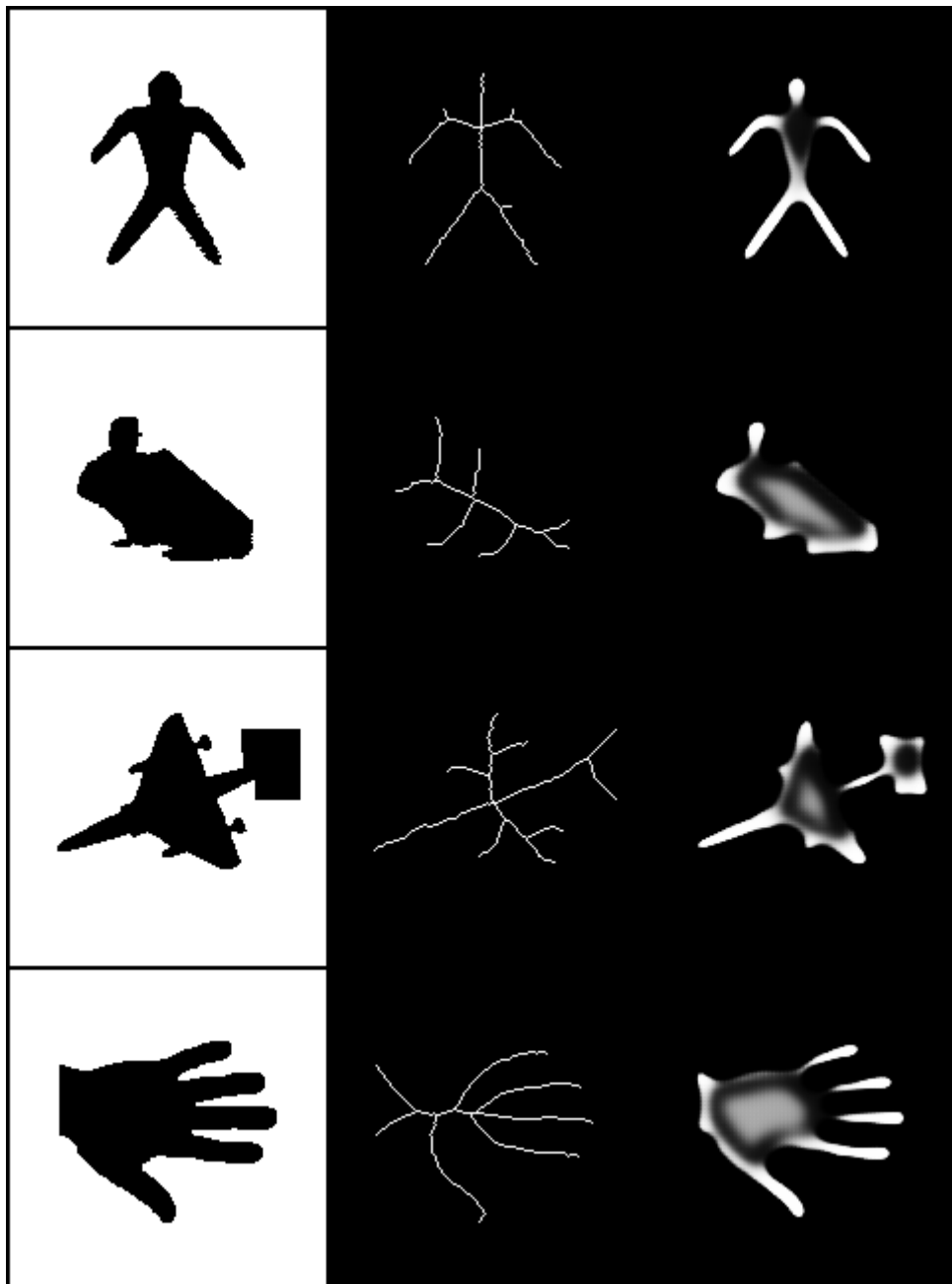


Figure 4.5 U-Net with Dice Loss trained on the Kimia216 dataset 10 epochs into the training inference on the validation dataset. (Left: original shape, Middle: ground truth mask, Right: model prediction)

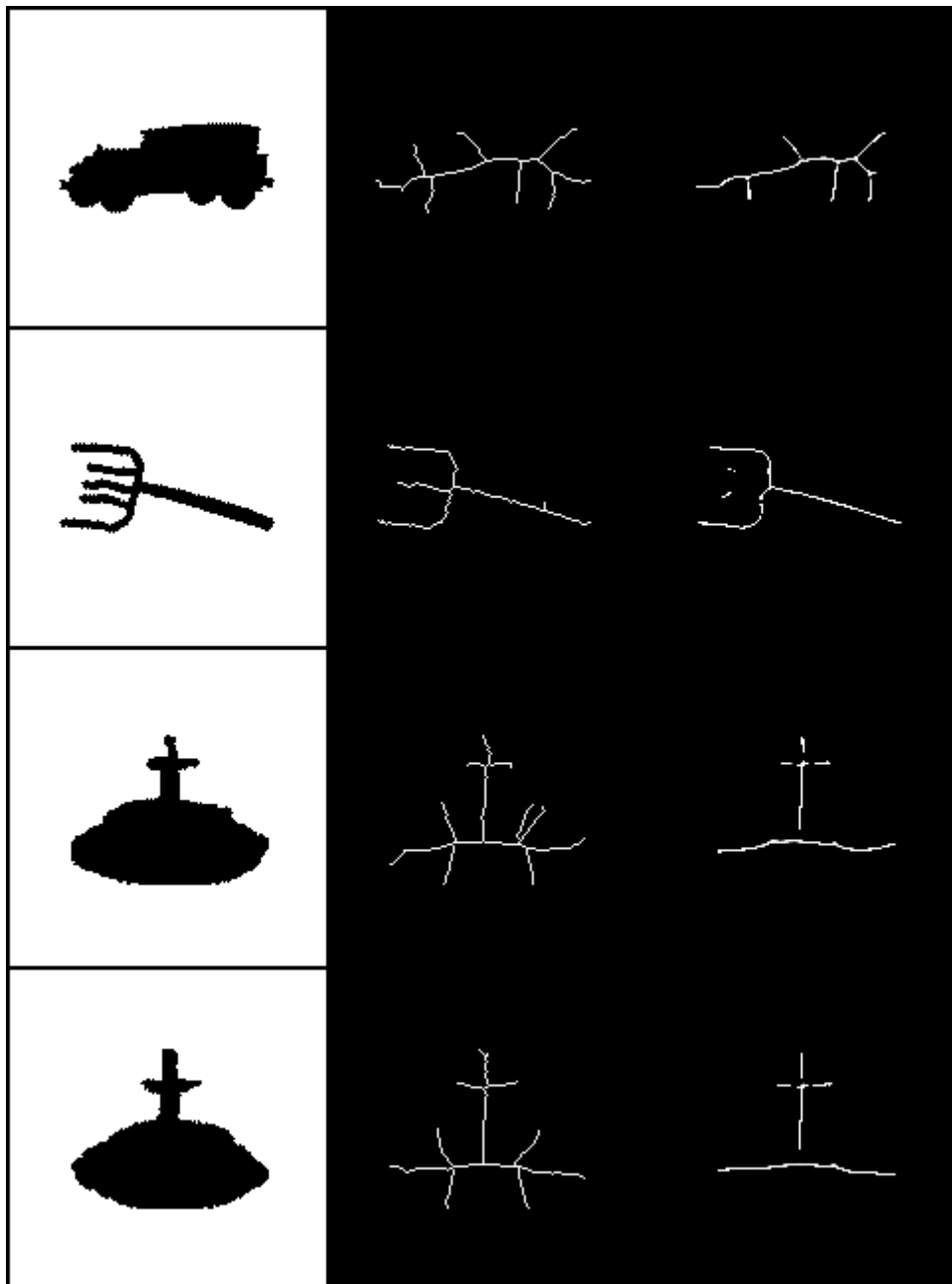


Figure 4.6 U-Net with Dice loss training on Kimia216 at epoch 85. (Left: original shape, Middle: ground truth mask, Right: model prediction)

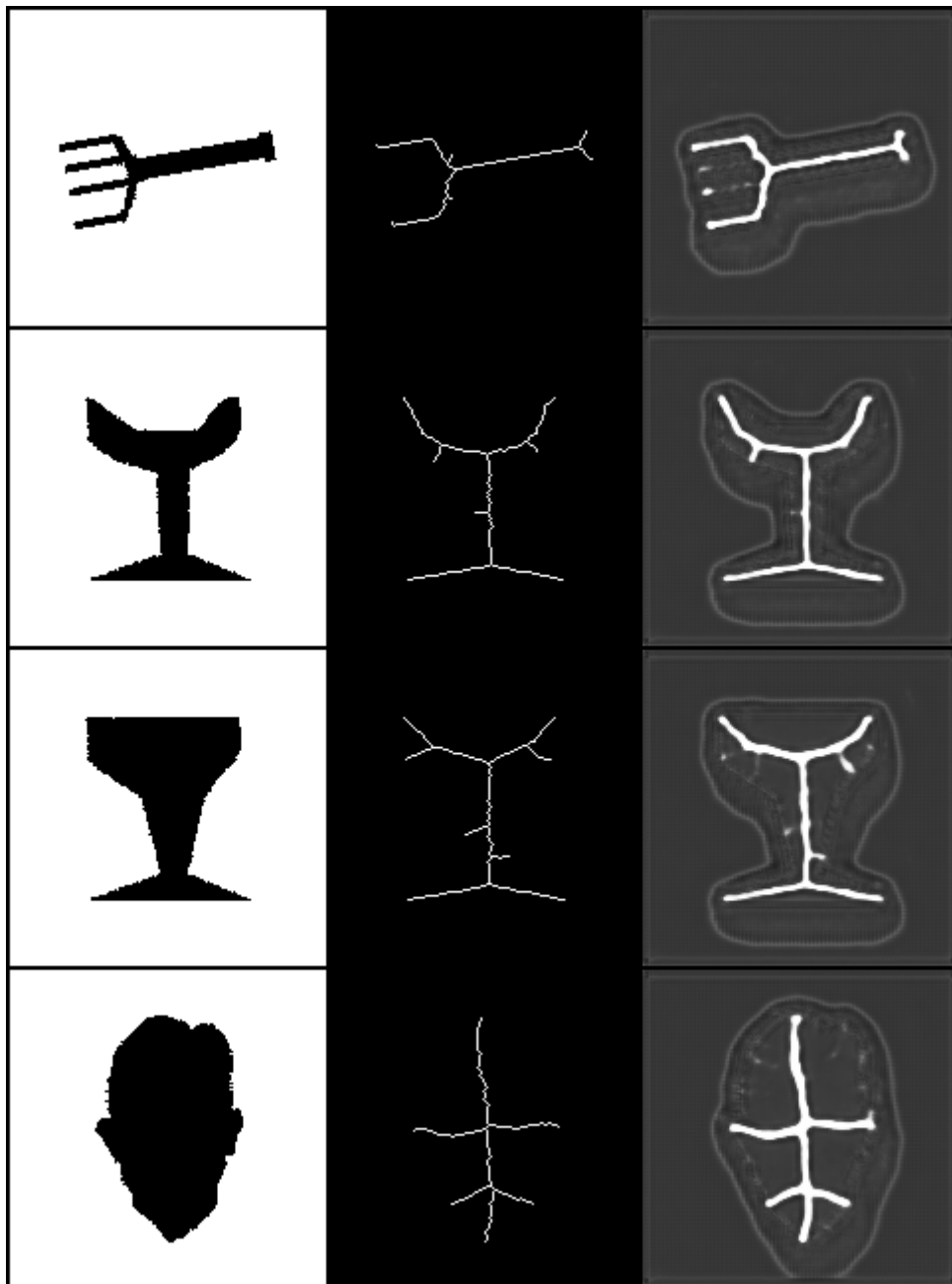


Figure 4.7 BU-Net with Dice loss training on Kimia216 at epoch 10. (Left: original shape, Middle: ground truth mask, Right: model prediction)

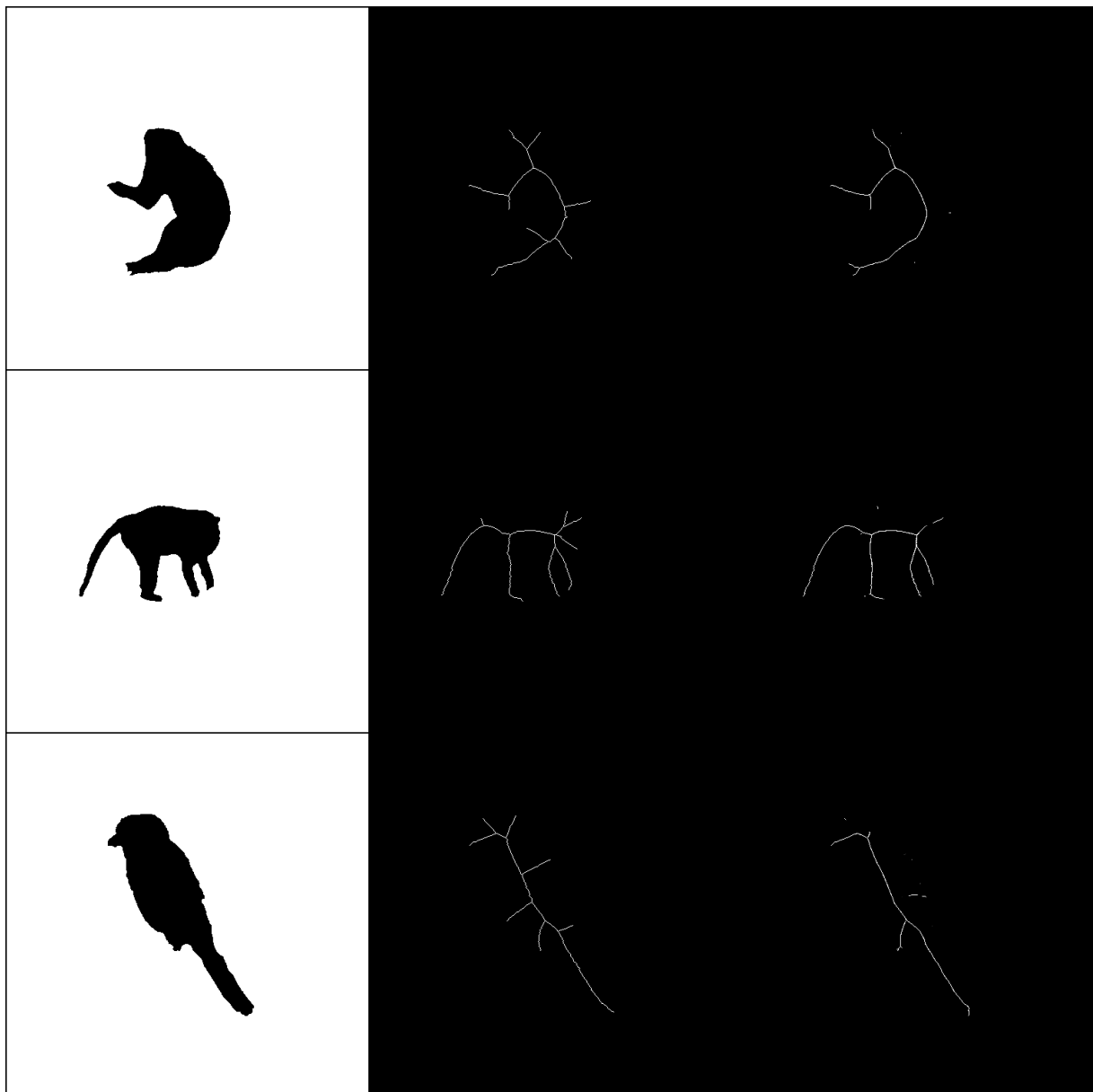


Figure 4.7 RAU-Net trained and tested on Animal2000. The images show the model's performance on unseen images from the same dataset. (Left: original shape, Middle: ground truth mask, Right: model prediction)

Magyar nyelvű összefoglaló

A vázkijelölés olyan folyamat melynek célja megtalálni egy jól határolt objektum közép-vonalát (Menten et al. 2023; Vargas et al. 2024, 257). Fontos, hogy ezzel gyakorlatilag az objektum legtömörebb formáját vesszük, amely még mindig ugyanazokkal a globális, lokális geometriai és topológiai jellemzőkkel rendelkezik (Ghanem and Anani 2021, 2). A hagyományok vázkijelölő algoritmusok kézi jellemzőkön, morfológiai technikákon és geometriai kapcsolaton alapulnak (Nyugen et al 2021, 1). Ezek a klasszikus technikák azonban igen érzékenyek az objektum határán található zajra és a legkisebb változások a formában vagy konturjában is gyakran olyan vázakat eredményeznek, amelyeknek túl sok vagy kevés ága van (Nguyen 2021, 106). Továbbá a hagyományos algoritmusoknak nincsen szemantikai ismerete ahhoz, hogy képesek legyenek vázkijelölésre természetes több objektumot tartalmazó képeken (Jerriothula et al. 2021, 14).

Mindeközben a mélytanulási képfeldolgozási módszerek egyre inkább megbízhatóvá váltak. Az elmúlt években nagy fejlődésen ment keresztül a szemantikus szegmentáció. Mivel a vázkijelölés tulajdonképpen a szemantikus szegmentáció alproblémájaképp kezelhető (tekintve, hogy gyakran úgy modellezik, mint egyfajta pixel pontos bináris osztályozási problémát, más szavakkal szemantikus szegmentációt (Nyugen et al 2021, 107)), ezért felvetül a kérdés hogyan lehetne a szemantikus szegmentációs megoldásokat adaptálni a vázkijelölés specifikusabb problémájára. Ez esetben a modellnek azt kell meghatároznia minden pixelre, hogy azok a vázhoz tartoznak vagy sem (Ko et al 2021, 89). Azonban ez a megközelítés is küzd néhány fontos kihívással: hogyan kényszerítsünk egy neurális hálót, hogy pontosan egy pixel vastagságú vázat jelöljön ki (Ko et al 2021, 80) és megőrizze az objektum topológiáját és a váz folytonosságát? Továbbá kihívást jelent a megfelelő metrikák hiánya, amelyekkel meglehetősen határozni a modellek teljesítményét a tanítás alatt (veszteségfüggvények) és a kiértékeléskor (pontossági metrikák).

A dolgozat célja, hogy felmérje a mélytanulási algoritmusok képességeit a bináris képek vázkijelölésén. Tekintettel a dolgozat kereteire és a hardver limitációkra csupán kis felbontású (160x160, 512x512) bináris képeken történik tanítás és értékelés, azonban fontos megjegyezni, hogy számos korszerű modell létezik vázkijelölésre zajos, természetes képeken is, amelyekre előfeldolgozás nélkül a tradicionális algoritmusok nem képesek.

A dolgozathoz készült modellek alapját a U-Net architektúra és annak különböző variánsai adják. Azért választottam a U-Net architektúrát, mert jelenleg ez az egyik legelterjedtebb modell a szemantikus szegmentációhoz, illetve alapvetően a bemeneti képek

valamely leredukált változatának visszaadása a feladata, amely tulajdonképpen a vázkijelölés feladatát is leírja (Ghanem 2021, 1).

A U-Net modelleken kívül különböző veszteségfunkciókat is teszteltek. A vázkijelöléshez megfelelő veszteségfunkciót találni ugyanis trükkös feladat három fontos kihívásnak köszönhetően: a vázkijelölésnél hatalmas az osztály-kiegyensúlyozatlanság (a váz és a háttérpixelék között); fontos a topológia megőrzése, illetve az egy pixel vastag folytonos váz. Ezért a dolgozat célja tesztelni hogyan képesek, illetve képesek-e a különböző veszteségfüggvények átlépni ezeken a kihívásokon.

Mivel a vázkijelölés tehát tekinthető a szemantikus szegmentáció egyik alproblémájának itt következik egy rövid irodalmi áttekintés olyan mélytanulási megoldásokról amelyek a szemantikus szegmentációt igyekeznek megoldani, illetve olyan megoldásokról is amelyek kifejezetten a vázkijelöléssel foglalkoznak. Ezek segítségével megérthetjük a különböző megoldások előnyeit és kihívásait.

A legfontosabb áttörés Long és társainak 2015-ös tanulmányából származik: ez volt az első teljesen konvolúciós neuronháló (Fully Convolutional Network, FCN) amely lehetővé tette a pixel-szintű osztályozást. A U-Net modell ennek a kiterjesztése volt azzal a céllal, hogy kevesebb tanító példa is elég legyen azonos szegmentációs eredmények mellett (Ronneberger és ts. 2015, 2). A U-Net architektúra számos korszerű megoldás alapjává vált.

Azonban a U-Net modellnek van néhány fontos korlátja: a háló mélysége limitált, mivel ha a háló túl mély a gradiensok hajlamosak „eltűnni” (vanishing gradients) (Li és ts. 2022, 4) illetve ez az architektúra még mindig nem képes a globális kontextus megőrzésére, ezért gyakran képtelen komplex képeket szegmentálni (Petit és ts. 2021, 2), továbbá a U-Net modellnek kihívást jelentenek a hatalmas osztály-kiegyensúlyozatlansággal (extreme class-imbalance) rendelkező feladatok, mint amilyen a vázkijelölés is.

Ugyanabban az évben, amikor a U-Net-et publikálták (2015) a Google egyik kutató csoportja nyilvánosságra hozott egy másik fontos architektúra megoldást: ResNet, amely úgynevezett „shortcut” kapcsolatokat használ arra, hogy megoldja a degradáció problémáját (ami azt jelenti, hogy ahogy a modell mélysége nő csökken a tanulási képessége) (He és ts. 2015, 8).

A U-Net koncepciói (az U alakú architektúra és az úgy nevezett „skip” kapcsolatok); a ResNet „shortcut” kapcsolatai alapvető megoldásokká váltak, amelyek együttesen formálták a mélytanulósos képfeldolgozás világát. Ezen koncepciók együttes alkalmazása egy új irányt határozott meg a szegmentációs kutatásokban.

Egy másik megoldást képviselnek az úgy nevezett „figyelem mechanizmusok” (attention mechanism). A figyelem mechanizmusokat azért vezették be a szegmentációs feladatokba, hogy csökkenteni próbálják a számítási igényeket és ne legyen szükség úgy nevezett „multi-stage” folyamatokra a feladatok megoldásához. „Figyelem kapukat” (Attention Gates vagyis AGs) lettek bevezetve, hogy megpróbálják növelni a modellek teljesítményét azzal, hogy ezek a kapuk automatikusan elnyomják a kép irreleváns részeit és kihangsúlyozzák a fontos objektumokat, anélkül, hogy növelnék a számítási igényeket (Oktay és ts. 2018, 1-2), ezzel eliminálva a külső lokalizációs modellek szükségességét (Oktay és ts. 2018, 8). Azonban még ezek a modellek sem voltak képesek 0.62 és 0.68 közöttinél jobb eredményt elérni az F1 metrikán – amely egy gyakran használt metrika a szegmentációnál, erről bővebben a következő fejezetekben lesz szó – vázkijelölés esetén, mivel ez az architektúra nem lett magára a vázkijelölés feladatára optimalizálva (Nguyen 2021, 2108). Ahhoz, hogy ezeket a korlátokat feloldják a figyelem mechanizmusok különböző más megoldásokkal lettek kombinálva.

Ugyan más egyéb említendő megoldások is készültek a vázkijelölésre mélytanulással, azonban ezek összefoglalása nem célja ennek a magyarnyelvű összefoglalónak, részletek ezekről a releváns fejezetben (Literature Review) találhatóak. A dolgozat alapvetően a három korábban tárgyalt mechanizmus, a U-Net architektúra, illetve a különböző veszteségfunkciók hatékonyságát méri a vázkijelölés feladatánál. Ehhez egy moduláris U-Net modellt készült (Residual Attention U-Net – RAU-Net), amelyben a különböző mechanizmusok egyenként kikapcsolhatóak, így létrehozva három további különböző modellre bontható: BU-Net (U-Net batch normalizációval, ez a mechanizmus nem kikapcsolható), AU-Net (Attention U-Net, amely az előbbi mellett figyelem mechanizmust is tartalmaz), RU-Net (Residual U-Net, amely a figyelem mechanizmus helyett reziduális blokkokat alkalmaz). Az implementáció és az architektúra részletei és diagramjai megtalálhatóak a dolgozatban.

A veszteségfunkciók és a metrikák témája egy szintén fontos probléma. Jelenleg nincsen olyan sztenderd mérőszám, amellyel valóban pontosan mérhető lenne a vázkijelölési teljesítményt, a kutatásokat amelyeket a dolgozat készítéséhez elolvastam több különféle metrikát használnak a teljesítmény meghatározására. Az egyik legelterjedtebb metrika a Dice mérőszám (matematikailag megegyezik a F1 mérőszámmal), amely úgynevezett területalapú metrika, vagyis azt méri mekkora az átfedés a modell által predikált és az elvárt osztályok között, hasonló metrikák még a Jaccard-Index avagy IoU (Intersection over Union). Ugyanakkor ezek a metrikák nem mérik a vázkijelölés fontos kritériumait: a topológia megőrzését és azt, hogy a váz egy pixel széles és töredezésektől mentes. Ezért a vázkijelölésnél fontos a kvalitatív vizuális analízis.

A dolgozat több veszteségfüggvényt is megvizsgál: a Dice veszteségfunkciót (amely a Dice mérőszám veszteségfunkciójából lett megalkotva) és azok kombinációt különböző más veszteségfüggvényekkel: DiceCE (Dice veszteségfüggvény kombinációja a kereszt-entrópia veszteségfüggvénnyel (Cross Entropy Loss), illetve az úgy nevezett Focal veszteségfüggvény kombinációját a Dice függvénnyel (Focal Dice Loss) és a Dice egy kifejezetten topológiára képzett variációja a cIDice metrika (centerline Dice).

A dolgozatban szereplő összes mélytanulási kísérlet egy Asus ROG Strix G16-os laptopon történt aminek Intel i7-13650HX-es a processzora, Windows 11-et futtat, illetve NVIDIA GeForce 4060 laptop GPU-t használ 8GB VRAM-mal. Mivel egyetlen laptop GPU-n történtek a kísérletek és a kutatás, ezért a hardver erőforrások jelentősen limitáltak voltak.

A modellek és a kísérletekhez szükséges kód a Python szkriptnyelv mélytanuláshoz népszerű PyTorch, TensorBoard könyvtárakkal illetve a kifejezetten szegmentációhoz kifejlesztett MONAI könyvtárral (Cardoso és ts. 2021) lett készítve. Azért választottam ezeket a könyvtárakat az implementációhoz, mert ezek hatékony implementációkat tartalmaznak a szükséges funkciókhoz.

A tanításhoz és teljesítmények méréséhez használt adatok a Yang és társai (2024) által készített SkeView nyilvános adathalmaz kollekciójából származnak. Ez az adatbázis azért lett használva, mert szemben a dolgozat korai szakaszaiban próbált generált adatbázissal, a SkeView adathalmaz validált jó minőségű nyilvános adathalmazokat tartalmaz, így könnyebbé téve a tanítást és az értékelést. A kollekcióból a Kimia99, Kimia216, MPEG400 és az Animal2000 adathalmazok lettek kiválasztva a kísérletekre, mivel ezekben igen sokféle objektum megtalálható. Sajnos a hardver limitációk miatt korlátozni kellett a képek méretét a Kimia adathalmazok esetén maximum 160x160-as felbontú, míg a többi adathalmaz esetén 512x512-es felbontású képek lettek felhasználva. Ezek a felbontások úgy lettek kiválasztva, hogy a lehető legtöbb kép felhasználásra kerüljön egy adatbázisból, illetve a Kimia adathalmazok esetén azzal a céllal is lett ilyen kicsire korlátozva a felbontás, hogy vizsgálható legyen hogyan teljesítenek a képek kisebb felbontású képeken tanítva. A méretkorlátok és a pontos alkalmazott kép-transzformációkról, illetve a kísérletek teljes háttéréről bővebben a dolgozat Experiments című fejezetében lehet olvasni.

A modellek eredményei javultak minél nagyobb adathalmazon voltak tanítva. A RU-Net és a BU-Net architektúrák olyan modelljei amelyek az Animal2000 adathalmazon voltak tanítva érték el a legjobb eredményeket. A RU-Net modell érte el a legjobb eredményt az Animal2000-ből vett teszt adathalmazon (0.826 Dice score). Az egyszerűbb BU-Net viszont

általánosságban jobb eredményeket ért el azokon az adathalmazokon amelyeken nem volt tanítva, tehát jobb általánosítási képességet mutatott. A legrosszabb eredményeket a mindhárom mechanizmus ötvözését tartalmazó RAU-Net model érte el (habár még ez is túlteljesítette a U-Net modellt által elért alapvető szintet). Ez azt mutatja, hogy a figyelmi mechanizmus ötvözése a reziduális tanulással valószínűleg kontraproduktív bináris képek vázkijelölése esetén.

A veszteségfüggvények teljesítménye kapcsán fontos megnézni mind a mérőszámokat, mind magukat a modellek által kijelölt vázakat. A Dice mérőszámban a Dice és DiceCE veszteségfüggvények gyakran jobban teljesítettek mint a Focal Dice veszteség függvény (FDL), azonban a nagyobb Animal2000 adathalmazokon tanított legtöbb modellen látszik (kivétel az AU-Net modellt), hogy az Focal Dice veszteségfüggvény használata kevesebb tört és zajos vázhoz vezet. Ennek oka látszik a tanítás során mentett model predikciókon, melyekből látszik, hogy a Focal Dice veszteségfüggvény már a tanulás korai szakaszában elkezd felismerni a vázak legfontosabb pontjait. Ez a kisebb adathalmazokon nem jár előnnyel, azonban a nagyobb Animal2000 adathalmaz esetén már látható különbséget hoz. Ez igazolja, hogy nagyobb adathalmazokkal, vagy több adathalmazon való tanítással tovább lehet növelni a modellek teljesítményét, továbbá, hogy a Dice mérőszám nem megfelelő a vázkijelölés értékelésére. Egy vázkijelölésre alkalmas mérőszám létrehozása arra is lehetőséget teremteni, hogy ezt veszteségfüggvénnyé adaptálják, amely növelné annak a lehetőségét, hogy mélytanulással meg lehessen oldani a vázkijelölés problémáját.

Összeségében a vázkijelölés mélytanulással továbbra is egy kihívást jelentő feladat. Habár az U-Net továbbfejlesztett változatai jobban teljesítettek, mint a klasszikus U-Net modell, azonban ezek a modellek sem tudtak olyan vázat kijelölni, amely megfelelt volna a vázkijelölés szigorú topológia feltételeinek. Nagy kihívást jelent a területen a megfelelő metrikák, veszteségfüggvények és nagy publikus adathalmazok hiánya, amelyeket a jövőbeli kutatások tudnak majd orvosolni.